

AMAZON SALES DATA ANALYSIS

Using SQL — Project Presentation

Sales performance across Yangon, Mandalay & Naypyitaw · Q1 2019



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Project Overview

1,000 transactions

across 3 branches (A, B, C) in Yangon, Mandalay, and Naypyitaw, Myanmar.

17 columns

covering invoice details, product line, pricing, tax, payment method, and customer ratings.

Time period: January – March 2019

The goal is to understand sales performance, customer behavior, product trends, and revenue contribution using SQL.

1,000

Total Records

3

Cities / Branches

6

Product Lines

Business Objectives

- 1 Analyze city-wise and branch-wise sales performance
- 2 Identify top-performing and low-performing product lines
- 3 Study customer behavior based on gender and customer type
- 4 Analyze sales trends over time (monthly, daily, hourly)
- 5 Calculate total revenue, COGS, and gross income
- 6 Identify preferred payment methods
- 7 Derive key business insights for decision-making

Tools & Technologies Used



MySQL

Database engine used to store and query the sales dataset



MySQL Workbench

GUI used for writing queries, importing data, and visual exploration



SQL

Core language for data wrangling, feature engineering, and analysis



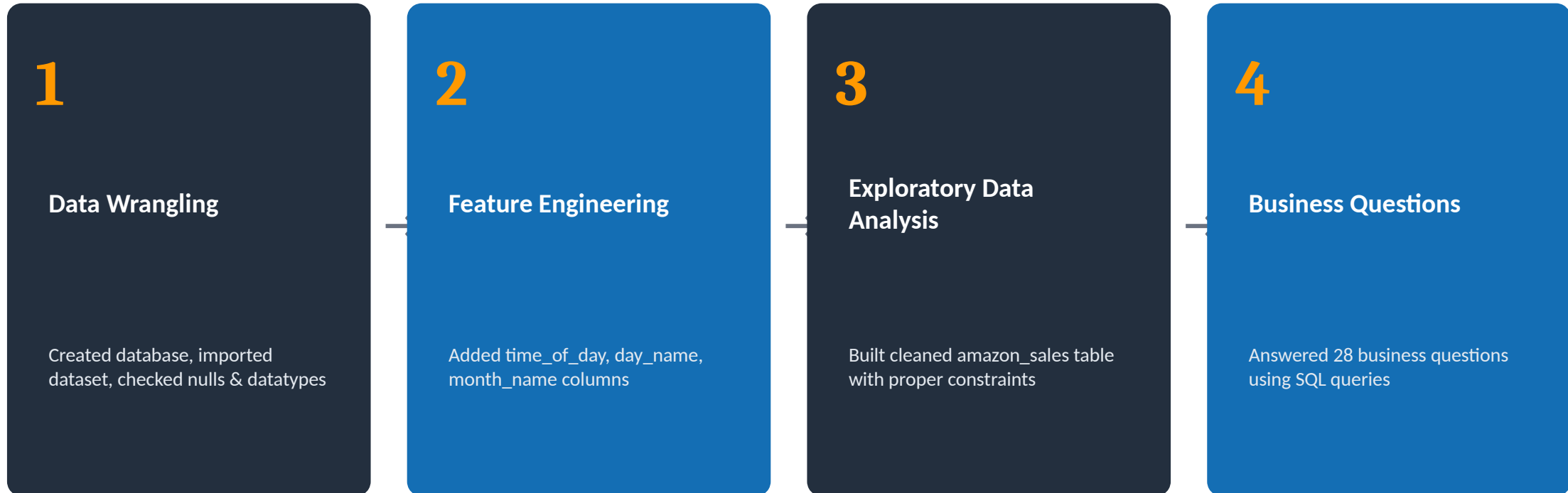
Python (Pandas & Matplotlib)

Used for supplementary data visualization (bar, pie, line charts)

Dataset Description

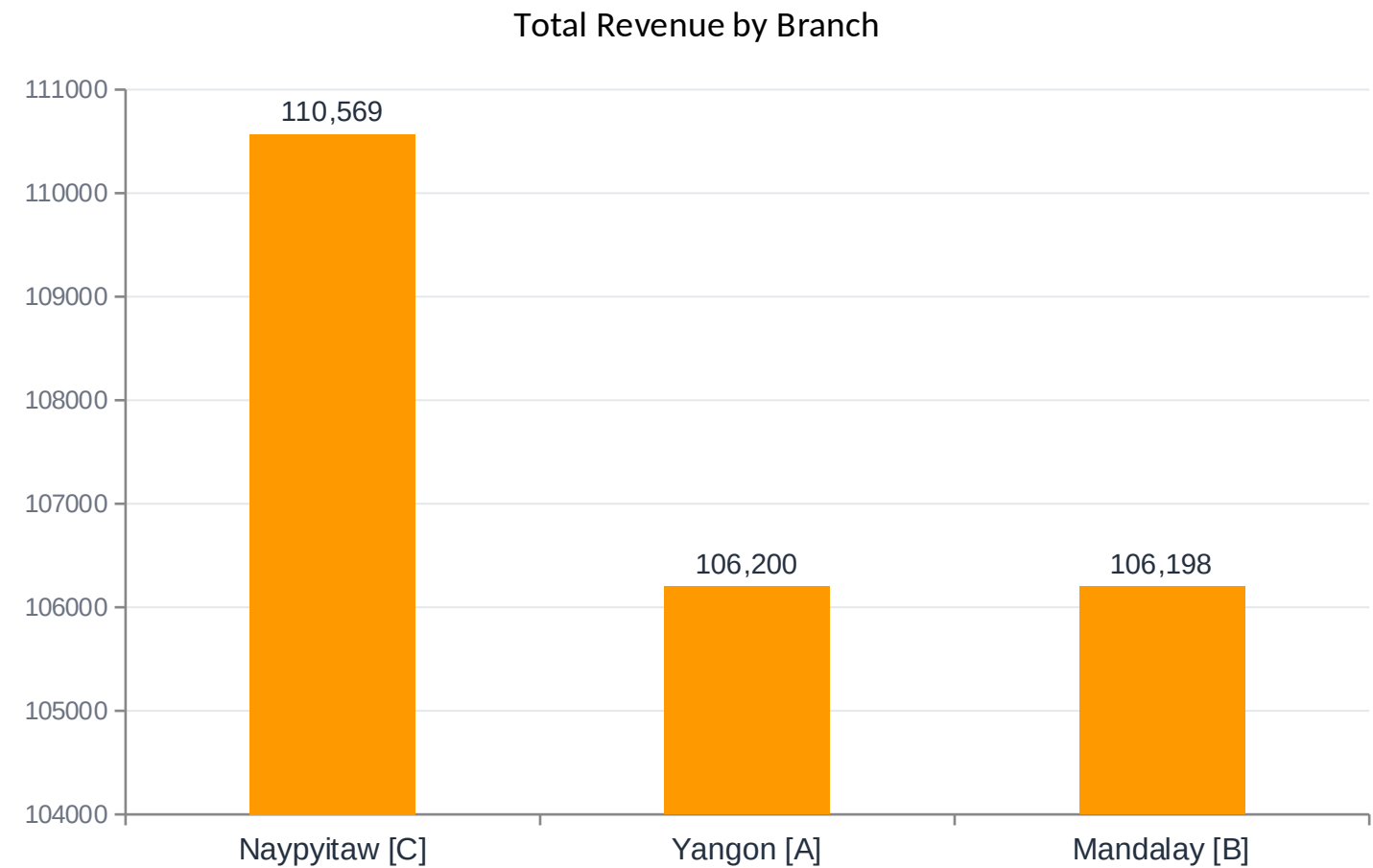
Column	Description
Invoice ID	Unique transaction identifier
Branch / City	Store branch (A, B, C) and its city
Customer type	Member or Normal
Gender	Customer gender
Product line	Product category (6 categories)
Unit price / Quantity	Price per unit and units purchased
Tax (5%) / Total	VAT applied and total bill amount
Date / Time	Transaction date and time
Payment	Cash, Credit card, or Ewallet
COGS / Gross income	Cost of goods sold and profit earned
Rating	Customer satisfaction rating (1-10)

Project Workflow



Revenue by City & Branch

Business Question: Which city / branch generates the highest revenue?



Top Performer

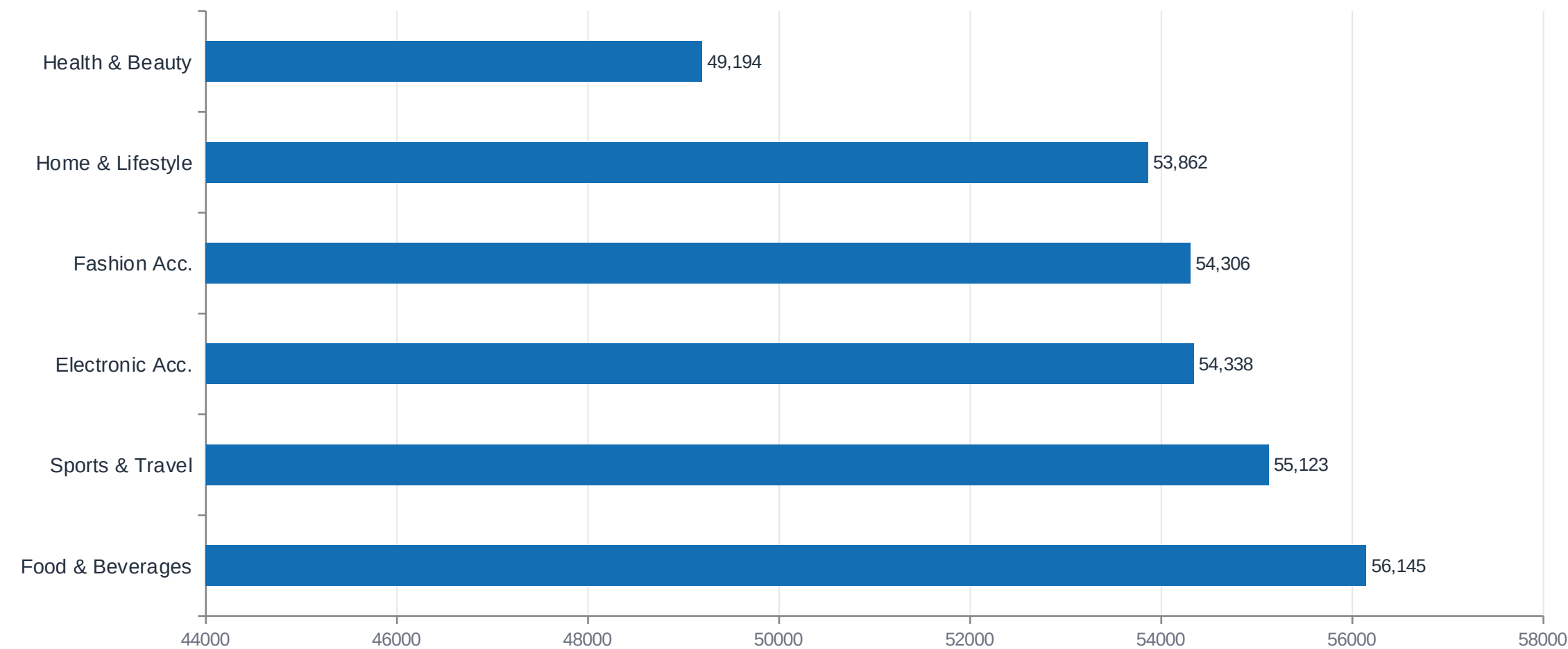
Naypyitaw

Branch C • \$110,568.71

Yangon and Mandalay follow closely, with revenue within \$4,400 of each other — showing balanced performance across the other two branches.

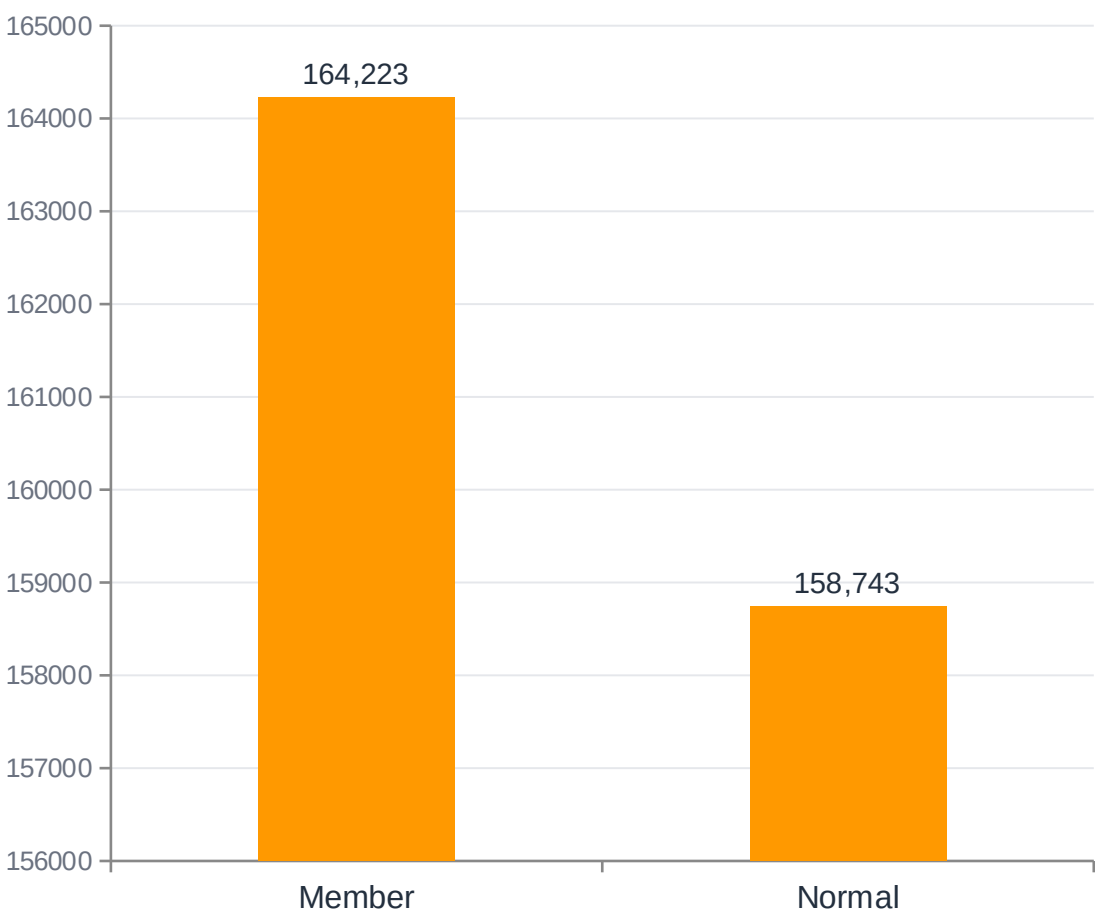
Revenue by Product Line

Business Question: Which product line generates the highest revenue?

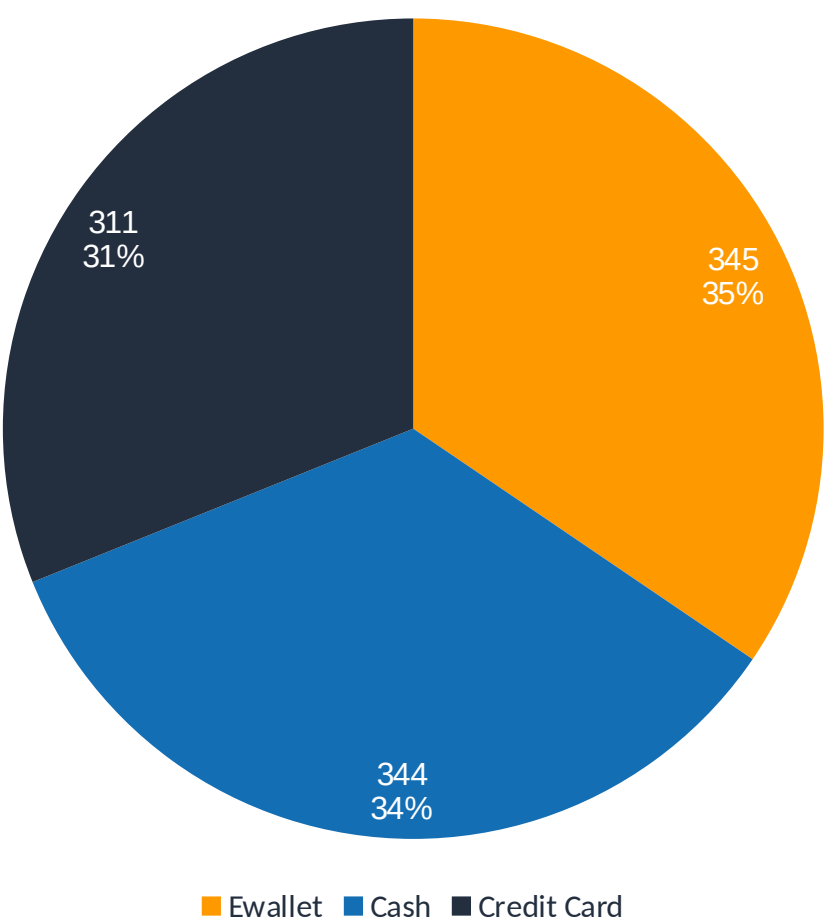


Customer & Payment Insights

Revenue by Customer Type

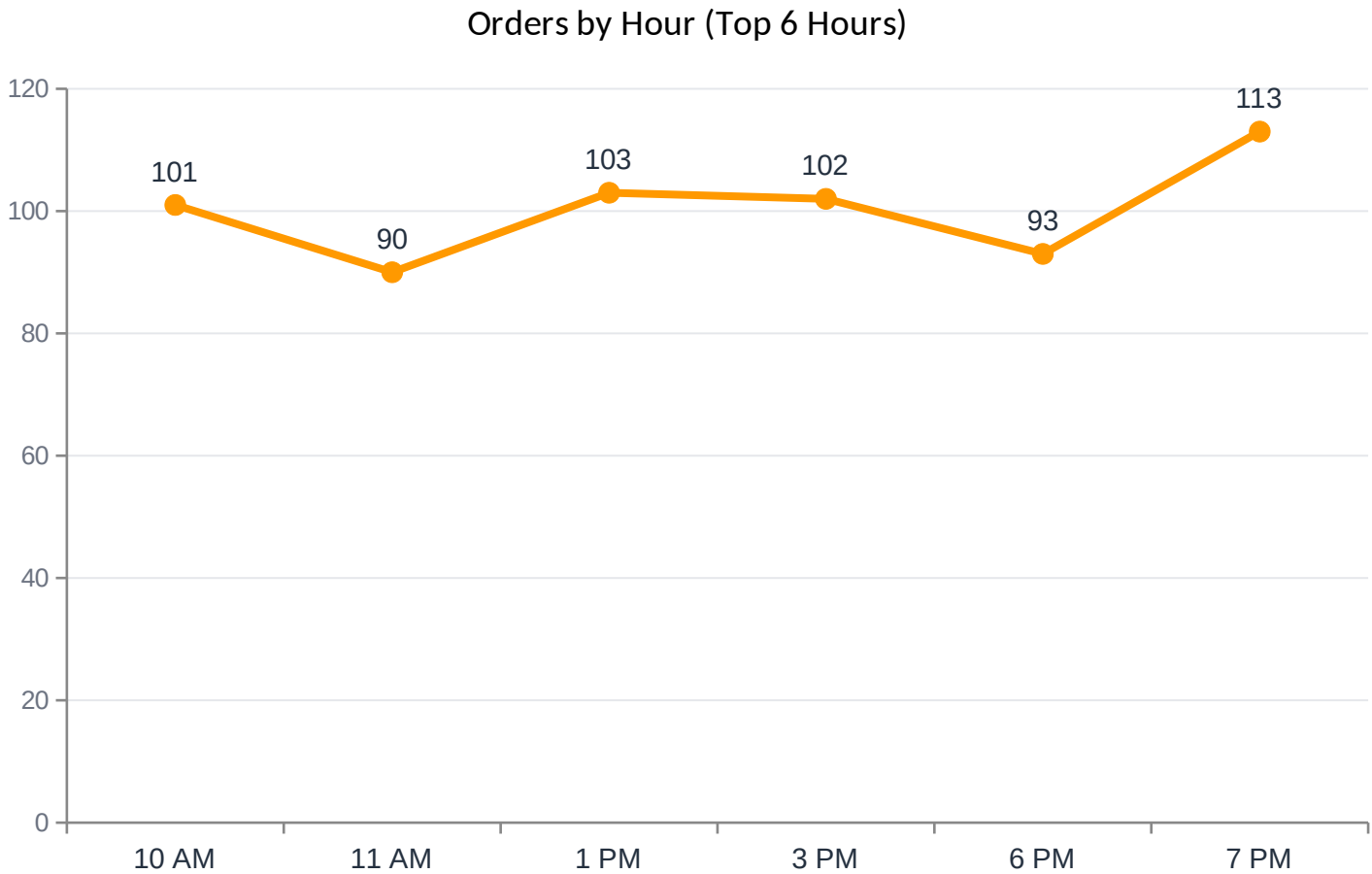


Payment Method Preference



Time-Based Sales Analysis

Business Question: At what time do customers shop the most?



Peak Sales Time

Afternoon

Peak Sales Day

Saturday

7 PM sees the single highest order count (113), suggesting strong evening footfall alongside the afternoon peak.

Key Findings & Insights



Product Analysis

- Electronic Accessories: highest unit sales (971 units)
- Food & Beverages: highest revenue (\$56,144.84)
- Health & Beauty: lowest sales and revenue



Sales Analysis

- January: highest revenue month
- Naypyitaw (Branch C): highest revenue city
- Afternoon & Saturday: peak sales periods



Customer Analysis

- Female customers contribute higher revenue
- Members outspend Normal customers
- Ewallet is the most preferred payment method

Conclusion & Future Scope

This project successfully analyzed Amazon sales data using SQL. By performing product, sales, customer, and time-based analysis, meaningful insights were derived — helping optimize product strategy, improve customer targeting, and enhance overall sales performance.

Future Scope

- Integration with Power BI or Tableau dashboards
- Advanced analysis using CTEs and window functions
- Predictive analysis using historical sales data
- Automated reporting pipelines